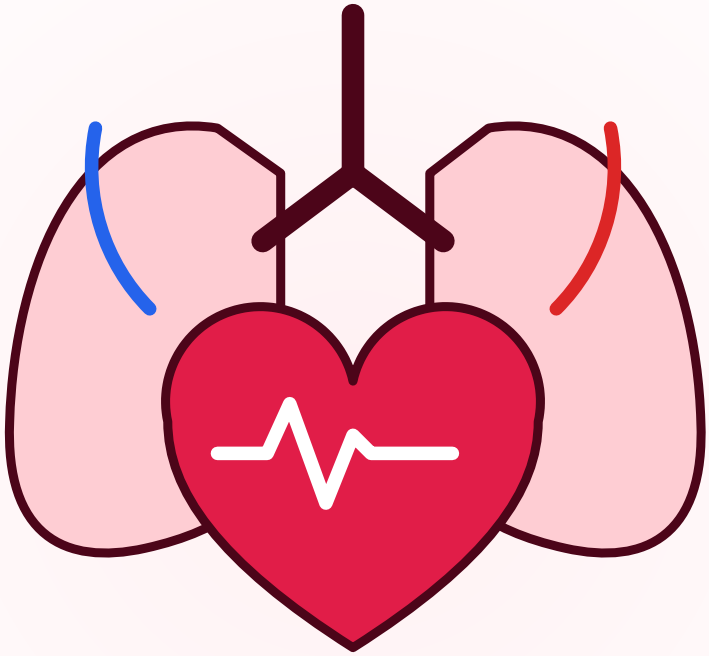
SECRET AGENT

# Case File

Every beat. Every breath. Every clue.



AGENT ID

**Name:** _____

**Partner:** _____

**Event:** Pump It Up

If found, please return to the DDMES Science Olympiad binder shelf.

# How a body agent works

Your heart, blood, and lungs are a secret delivery team. The lungs bring in **oxygen**, the heart pumps it all over the body, and the blood carries away **carbon dioxide**. Your job: learn every part so well you can solve any case with no notes. This case file lives in your SciOly binder.

**1**
**LEARN**
Know the heart, lung, and blood parts.

**2**
**TRACE**
Follow the blood and the air, step by step.

**3**
**MODEL**
Use models: balloons, grapes, straws, a floor heart.

**4**
**MEASURE**
Count pulse and breaths like a scientist.

**5**
**SOLVE**
Name the condition from the clues.

## The Agent's Code

- [ ] Practice with no notes, like the tournament.
- [ ] Find a pulse with 2 fingers, gently. Never use your thumb.
- [ ] Exercise only as hard as feels okay.
- [ ] Never share or use someone else's medicine or inhaler.
- [ ] Health conditions are nobody's fault. Talk about them kindly.
- [ ] Face drooping or chest pain in real life? Tell an adult and call 911.

## My agent kit

- [ ] This case file and 2 sharp pencils
- [ ] Red and blue colored pencils
- [ ] A watch or timer with seconds
- [ ] A measuring tape (centimeters)
- [ ] Your flashcards for all 4 lessons
- [ ] A family member to measure

## Journal map


| | |
|---|---|
| Know your event | p. 3-7 |
| Cases #1 to #4 | p. 8-11 |
| Station Drill | p. 12 |
| Logs and trackers | p. 13-14 |
| Missions and badges | p. 15-16 |
| Answer key (upside down) | p. 17 |


## Where to look

**Your wrist and neck**
pulse points: where you can feel an artery

**The grocery store**
grapes and broccoli look like parts of the lungs

**The garage or closet**
a vacuum hose has rings, like the trachea

**A bare tree in winter**
upside down, it looks like your airway tree

**The kitchen**
a sponge, a turkey baster, a squeeze bottle

**A doctor's office**
heart and lung posters on the wall

**Did you know?** Your heart pumps blood about 100,000 times a day, and your lungs have millions of tiny air sacs called alveoli.

# Mission briefing: the event

Pairs rotate through timed hands-on stations for up to 50 minutes (label diagrams, study models, answer scenario questions about the heart, blood, lungs, and diseases) and bring only pencils, no notes.

## How the tournament works

- [ ] You and your partner move through stations.
- [ ] When time is called, stop and rotate. No skipping ahead or going back.
- [ ] Bring only pencils: no notes, flashcards, or books.
- [ ] Our practice test: 50 minutes at 13 stations, about 3 1/2 minutes each.

## What stations look like

- [ ] Label a heart or airway diagram.
- [ ] Study a model (balloon lungs, blood in a jar, a floor heart).
- [ ] Read a scenario and name the condition.
- [ ] Read pulse and breathing data.

## Key words

| | |
|---|---|
| **Circulatory system** | Heart, blood, and blood vessels. It moves blood around the body. |
| **Respiratory system** | Nose, airways, and lungs. It brings air in and out. |
| **Oxygen** | The gas your cells need. You breathe it in. |
| **Carbon dioxide** | The waste gas your cells make. You breathe it out. |
| **Pulse** | The beat you feel in an artery each time the heart pumps. |

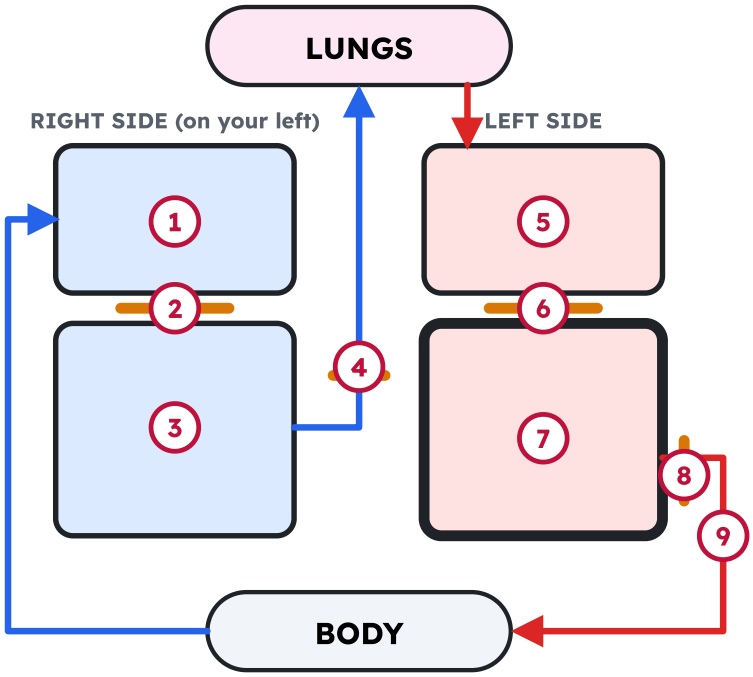
## What a Pump It Up agent can do

Check each one when you can do it with no notes.

- [ ] name the 4 heart chambers (right and left atria, right and left ventricles) and the aorta, and say what each does.
- [ ] name the 4 valves in order (tricuspid, pulmonary, mitral, aortic) and explain that valves are one-way doors.
- [ ] trace blood from the body, through the heart and lungs, and out the aorta.
- [ ] name the parts of blood (red cells, white cells, platelets, plasma) and their jobs.
- [ ] compare arteries, veins, and capillaries.
- [ ] put the airway in order: nose (or mouth), pharynx, larynx, trachea, bronchi, bronchioles, alveoli.
- [ ] explain the lungs, lobes, and diaphragm, and use a model to show breathing.
- [ ] explain how oxygen and carbon dioxide trade places in the lungs and how the two systems work together.
- [ ] measure pulse and breathing rate and explain why both go up with exercise.
- [ ] explain sickle cell anemia, heart attack, and stroke (organ affected, causes, prevention, treatment).
- [ ] explain asthma and cystic fibrosis (organ affected, causes, prevention, treatment).
- [ ] explain COPD, lung cancer, and congestive heart failure (organ affected, causes, prevention, treatment).
- [ ] tell which conditions are genetic and which come mostly from habits or the environment, and name healthy habits.
- [ ] answer station questions on diagrams, models, and scenarios with no notes.

# Inside the heart

Your heart is a muscle with **4 rooms** (chambers) and **4 one-way doors** (valves). The right side pumps oxygen-poor blood (**blue**) to the lungs. The left side pumps oxygen-rich blood (**red**) to the body.




| | |
|---|---|
| 1 | **Right atrium** |
| 2 | **Tricuspid valve** |
| 3 | **Right ventricle** |
| 4 | **Pulmonary valve** |
| 5 | **Left atrium** |
| 6 | **Mitral valve** |
| 7 | **Left ventricle** |
| 8 | **Aortic valve** |
| 9 | **Aorta** |

**Mirror trick:** the heart faces you, so its right side is on YOUR left.

The left ventricle has the thickest wall: it pumps blood to the whole body.

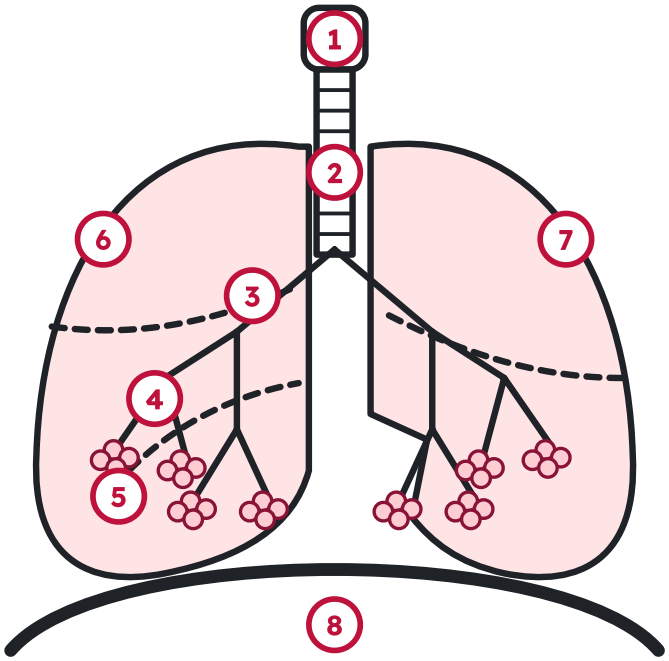
## Heart parts

| | |
|---|---|
| **Atria (atrium = one)** | The 2 top chambers. They receive blood coming into the heart. |
| **Ventricles** | The 2 bottom chambers. They pump blood out of the heart. |
| **Tricuspid valve** | Between the right atrium and right ventricle. |
| **Pulmonary valve** | Between the right ventricle and the artery to the lungs. |
| **Mitral valve** | Between the left atrium and left ventricle. |
| **Aortic valve** | Between the left ventricle and the aorta. |
| **Aorta** | The biggest artery. It carries oxygen-rich blood from the left ventricle to the body. |

**Path of blood:** Body > right atrium > tricuspid valve > right ventricle > pulmonary valve > lungs (drop off carbon dioxide, pick up oxygen) > left atrium > mitral valve > left ventricle > aortic valve > aorta > body. The “lub-dub” of a heartbeat is the valves snapping shut.

# The airway and lungs

Air takes the same trip every time: **nose (or mouth) > pharynx > larynx > trachea > bronchi > bronchioles > alveoli**.




| | |
|---|---|
| 1 | **larynx (voice box)** |
| 2 | **trachea (windpipe)** |
| 3 | **bronchi** |
| 4 | **bronchioles** |
| 5 | **alveoli** |
| 6 | **right lung: 3 lobes** |
| 7 | **left lung: 2 lobes** |
| 8 | **diaphragm** |

The left lung is smaller (2 lobes) to leave room for the heart.

## Parts of the respiratory system

| | |
|---|---|
| **Nose and nasal cavity** | Air comes in. Hairs and mucus trap dust; the air gets warm and wet. |
| **Mouth** | Another way for air to get in and out. |
| **Pharynx** | The throat. Air and food both pass through it. |
| **Larynx** | The voice box. Air passing through it lets you talk. |
| **Trachea** | The windpipe. Stiff rings keep it open. |
| **Bronchi** | Two tubes that split off the trachea, one into each lung. |
| **Bronchioles** | Smaller and smaller tubes branching inside the lungs. |
| **Alveoli** | Millions of tiny air sacs covered with capillaries. Gases trade places here. |
| **Lungs and lobes** | Two spongy organs. Right lung: 3 lobes. Left lung: 2 lobes. |
| **Diaphragm** | A dome-shaped muscle under the lungs. Down = breathe in. Up = breathe out. |

# Blood and blood vessels

Blood is a delivery service. It travels through tubes called **blood vessels**, all around your body and back.

## Parts of blood

| Part | What it is | Job |
|---|---|---|
| **Plasma** | The yellowish liquid part. About half of your blood. | Carries food, water, and wastes. |
| **Red blood cells** | Round disks, dented in the middle. | Carry oxygen. |
| **White blood cells** | Bigger cells, not as many. | Fight germs and infections. |
| **Platelets** | Tiny pieces of cells. | Make a clot to stop bleeding. |

## Blood vessels

| Vessel | Carries blood… | Walls and size |
|---|---|---|
| **Artery** | AWAY from the heart. You feel your pulse in one. | Thick, stretchy walls to handle the heart's strong push. |
| **Vein** | BACK to the heart. | Thinner walls, with little valves so blood can't flow backward. |
| **Capillary** | Between arteries and veins, right next to the cells. | The tiniest. Walls only one cell thick, so gases pass through. |

### Gas exchange

In the **alveoli**, oxygen moves from the air into the blood in the capillaries. Carbon dioxide moves from the blood into the air, and you breathe it out. The two systems work as a team!

### Pulse and breathing

Count beats for 15 seconds and multiply by 4. Count breaths for 30 seconds and multiply by 2.

Typical kids at rest: about 70 to 110 beats and 16 to 25 breaths per minute. Both go up with exercise, because working muscles need more oxygen.

**Blood model:** syrup = plasma, red candies = red blood cells, white beans = white blood cells, sprinkles = platelets.

# Case board: the 8 conditions

At the tournament you may be asked about all 8. For each one, know the organ it affects, the cause, prevention, and treatment. COPD = Chronic Obstructive Pulmonary Disease.

| Condition | Affects | Cause | Prevention | Treatment |
|---|---|---|---|---|
| **Sickle cell anemia** Genetic | Blood: red blood cells. | Genetic. A person gets the sickle cell gene from both parents. It is not catching. | It cannot be prevented, but babies are tested at birth. Drinking lots of water, resting, and staying warm help prevent pain episodes. | Medicines, blood transfusions, and for some people a bone marrow transplant. |
| **Heart attack** Habits or environment | The heart muscle and the arteries that feed it. | Smoking, high blood pressure, high cholesterol, diabetes, not enough exercise, family history. | Exercise, healthy food, no smoking, regular checkups. | Call 911 fast. Doctors use medicine, a tiny tube (stent) to open the artery, or surgery. |
| **Stroke** Habits or environment | The brain and the blood vessels that feed it. | High blood pressure, smoking, heart problems, diabetes. Sickle cell anemia can raise the risk in kids. | Same healthy habits as for the heart; keep blood pressure healthy. | Fast medicine to break up the clot or a procedure to remove it, then therapy to relearn skills. |
| **Congestive heart failure** Habits or environment | The heart. Fluid can back up into the lungs and legs. | A past heart attack, high blood pressure, or damaged valves. | Healthy food, exercise, no smoking, keep blood pressure healthy. | Medicines (including water pills to remove extra fluid), less salt, and sometimes devices or surgery. |
| **Asthma** A mix | Lungs: the airways (bronchi and bronchioles). | Often runs in families. Triggers include smoke, dust, pollen, pet dander, cold air, colds, and hard exercise. | You cannot always prevent having asthma, but you can prevent attacks: avoid triggers and take controller medicine. | Inhalers: a quick-relief (rescue) inhaler opens airways fast; a controller medicine lowers swelling. |
| **Cystic fibrosis (CF)** Genetic | Lungs and the digestive system (especially the pancreas). | Genetic. A child gets a CF gene from both parents. It is not catching. | It cannot be prevented. Babies are tested at birth so care can start early. | Daily airway clearing (special vest or chest tapping), inhaled medicines, enzyme pills with meals, and newer medicines that help the body work better. |
| **COPD** Habits or environment | Lungs: the airways and alveoli get damaged over many years. | Mostly smoking. Also breathing smoke, dust, or polluted air for a long time. | Do not smoke and stay away from smoke. | Quit smoking, inhalers, extra oxygen, and breathing exercise programs. |
| **Lung cancer** Habits or environment | Lungs. | Mostly smoking and secondhand smoke. Also radon (a gas from the ground), pollution, and sometimes genes. | Do not smoke, avoid smoke, test your home for radon. | Surgery, chemotherapy, radiation, and other medicines. |

**Not catching:** none of these spread from person to person like a cold. Genetic means you are born with it, from genes from both parents.

**Draw it: a normal red blood cell**

**Draw it: a sickle cell**

# Case #1: Family Pulse Check

TOP SECRET

**ASSIGNED** after Lesson 1 (Fri Oct 16) · **BADGE** **Pulse Detective**

**YOUR MISSION** Family pulse check (15 min). Teach someone at home to find a pulse. Measure it at rest and after 1 minute of marching; write both numbers on the back. Make 5 flashcards (circulatory system, respiratory system, oxygen, carbon dioxide, pulse) and quiz yourself twice.

## How to find a pulse

1. Put 2 fingers (not your thumb) on the thumb side of the wrist.
2. Press gently until you feel a beat.
3. Count beats for 15 seconds.
4. Multiply by 4 = beats per minute.

## Breaths

1. Watch the chest go up and down.
2. Count breaths for 30 seconds.
3. Multiply by 2 = breaths per minute.
4. Tip: people breathe differently when they know you're counting!

## Our data

| Person (me, mom, grandpa...) | Pulse at rest | Pulse after 1 min of marching | Breaths at rest | Breaths after marching |
|---|---|---|---|---|
| **Me** | | | | |
| | | | | |
| | | | | |
| | | | | |
| | | | | |

## Graph it

Draw 2 bars for each person: pulse at rest and after marching. Color them different colors.



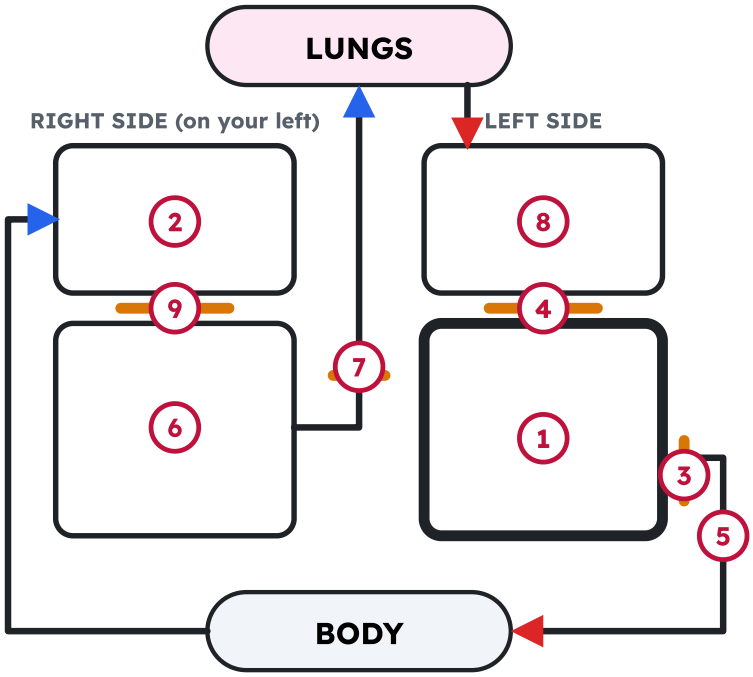

**Why did everyone's pulse and breathing go up at the same time?**

# Case #2: The Heart Map

TOP SECRET

**ASSIGNED** after Lesson 2 (Fri Nov 20) · **BADGE** ♡ **Heart Mapper** 💧 **Blood Tracer**

**YOUR MISSION** Someone scrambled the labels on this heart map. Name all 9 parts with no peeking at p. 4 (**Heart Mapper**). Then color it: oxygen-poor blood blue, oxygen-rich blood red, and trace the path of blood (**Blood Tracer**). Also make your 7 heart flashcards.




| | |
|---|---|
| 1 | |
| 2 | |
| 3 | |
| 4 | |
| 5 | |
| 6 | |
| 7 | |
| 8 | |
| 9 | |

## Trace the path of blood

Word bank: right atrium · left atrium · right ventricle · left ventricle · tricuspid · pulmonary · mitral · aortic · lungs · aorta

Body > _____ > _____ valve > _____ > _____ valve > _____ > _____ > _____ valve > _____ > _____ valve > _____ > Body

**Which side of the heart has oxygen-poor blood?**

**Why does blood go to the lungs between the two sides?**

**What do valves do?**

**Which chamber has the thickest wall? Why?**

# Case #3: The Model Hunt

TOP SECRET

**ASSIGNED** after Lesson 3 (Fri Jan 15) · **BADGE** **Model Maker**

**YOUR MISSION** Scientists use **models** to study things that are hard to see. Hunt around your house for things that work like body parts. Find at least 5 (the grapes are done for you). Also make your 7 blood and vessel flashcards.



Stems = bronchioles, grapes = alveoli. Photo: Photo: tonyrebelo, iNaturalist, CC-BY-SA

## Ideas to hunt for

A vacuum or dryer hose, broccoli, a bare tree branch, a kitchen sponge, a balloon, a straw or garden hose, a squeeze bottle or turkey baster, a pet door or a swinging door.

**Careful:** ask before you borrow anything, and never put a balloon or bag over your face.

## My model hunt

| Body part | Model I found at home | How is it like the body part? | ✓ |
|---|---|---|---|
| **Alveoli (air sacs)** | *grapes on a stem* | | |
| **Trachea with rings** | | | |
| **Airway tree (bronchi and bronchioles)** | | | |
| **Spongy lungs** | | | |
| **A lung filling with air** | | | |
| **A blood vessel** | | | |
| **The heart pumping** | | | |
| **A one-way valve** | | | |

**Pick one model. How is it DIFFERENT from the real body part?**

# Case #4: Mystery Patients

TOP SECRET

**ASSIGNED** after Lesson 4 (Fri Feb 12) · **BADGE** Case Solver

**YOUR MISSION** Six patients need an agent! Read each case file, name the condition, and write the organ it affects. No peeking at p. 7. Then do your final review: quiz your 2 weakest topics until you get them right 3 times.

| # | The case | Condition | Organ affected |
|---|---|---|---|
| 1 | A 9-year-old soccer player coughs and wheezes during practice on cold mornings. She uses an inhaler before games. | | |
| 2 | A grandfather mowing the lawn feels crushing chest pain. An artery to his heart muscle is blocked, and doctors open it with a stent. | | |
| 3 | A baby's newborn blood test shows red blood cells that can bend into a curved crescent shape. Both parents carry the gene. | | |
| 4 | A woman who smoked for 30 years breathes out slowly and gets tired walking to the mailbox. Her doctor gives her inhalers and extra oxygen. | | |
| 5 | A neighbor suddenly can't smile on one side of his face, his arm feels weak, and his words come out jumbled. | | |
| 6 | A boy does airway clearing with a shaking vest every morning and takes enzyme pills with his meals. | | |

## Genetic or not?

Which 2 of the 6 patients were born with their condition?

_____

Which patient should call 911 right away?

_____

## Final review

My 2 weakest topics: _____

and _____

Draw the heart path once more from memory:

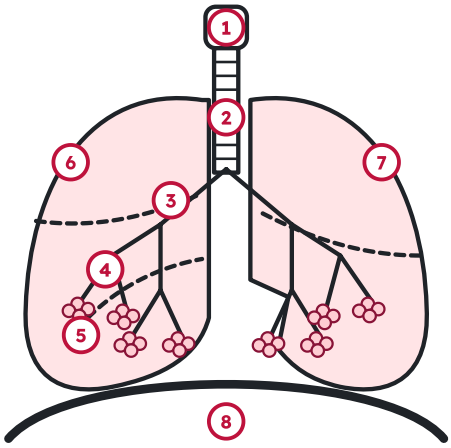
# Station Drill

NO NOTES

Set a timer for **3 1/2 minutes** per station, like the tournament. No notes! Check your answers on p. 17. Get Station 1 right for **Airway Explorer**, and all 5 stations right for **Station Ace**.

## Station 1: Airway

Name each numbered part.




| | |
|---|---|
| 1 | |
| 2 | |
| 3 | |
| 4 | |
| 5 | |
| 6 | |
| 7 | |
| 8 | |

**Put the air path in order, starting at the nose:** bronchi, larynx, alveoli, pharynx, trachea, bronchioles

## Station 2: Blood

1. Which part of blood carries oxygen? _____

2. You scrape your knee. What makes the clot? _____

3. Which part fights germs? _____

## Station 3: Vessels

4. Which vessels carry blood away from the heart? _____

5. Where do gases trade places with cells? _____

6. Which vessels have little valves inside? _____

## Station 4: Data

An agent counts 22 heartbeats in 15 seconds at rest. After 2 minutes of jumping jacks, she counts 35 beats in 15 seconds.

7. What is her resting pulse in beats per minute? Show your math. _____

8. What is her pulse after jumping jacks? _____

9. Why did it go up? _____

## Station 5: Team work

10. In 2 sentences, explain how the heart and lungs work as a team.

# Evidence: Practice Log

Log every time you practice stations: at meetings, the mock tournament (Feb 12), the practice test, or at home.

| Date | Topic or stations | Right / total | What I missed | Fixed it? (how) |
|---|---|---|---|---|
| | | | | |
| | | | | |
| | | | | |
| | | | | |
| | | | | |
| | | | | |
| | | | | |
| | | | | |
| | | | | |
| | | | | |
| | | | | |
| | | | | |
| | | | | |

## Flashcard tracker

Check a box each time you get the WHOLE set right. Get every set right twice for the **Flashcard Champ** badge.

| Flashcard set | Right once | Right twice | Date |
|---|---|---|---|
| **Lesson 1: 5 key words** | | | |
| **Lesson 2: 7 heart parts** | | | |
| **Lab I1: 10 airway parts** | | | |
| **Lesson 3: 7 blood and vessel words** | | | |
| **Lab I2: 4 heart and blood conditions** | | | |
| **Lab I3: 4 lung conditions** | | | |

# Healthy Habits Tracker

Heart attack, stroke, COPD, and lung cancer come mostly from habits and the environment. Healthy habits protect your heart and lungs for life! Track 7 days for the **Healthy Habits Hero** badge.

| Habit | Day 1 | Day 2 | Day 3 | Day 4 | Day 5 | Day 6 | Day 7 |
|---|---|---|---|---|---|---|---|
| **Moved or played hard for 60 minutes** | | | | | | | |
| **Drank water instead of soda** | | | | | | | |
| **Ate a fruit or vegetable** | | | | | | | |
| **Stayed away from smoke** | | | | | | | |
| **Got a good night's sleep** | | | | | | | |
| **Washed hands (fewer germs, fewer colds)** | | | | | | | |

## Which condition does each habit help prevent?

| Habit | Condition it helps prevent |
|---|---|
| Not smoking | |
| Exercise | |
| Healthy food with less salt | |
| Testing a home for radon | |

## Heart rate challenge

Take your pulse before and after your favorite exercise.

Exercise: _____

Before: _____ After: _____

## My healthy habits promise

_____

_____

_____

_____

# Case Missions

One case for each Pump It Up meeting. When you finish, show a coach or your Event Captain for initials, then color in the badge.

| When | Mission | Badge | Done | Initials |
|---|---|---|---|---|
| **After Fri Oct 16** Lesson 1 | Case #1: Family Pulse Check (p. 8). | Pulse Detective | ☐ | |
| **After Fri Nov 20** Lesson 2 | Case #2: label the Heart Map and trace the blood (p. 9). | Heart Mapper / Blood Tracer | ☐ | |
| **After Fri Jan 15** Lesson 3 | Case #3: find 5 body models at home (p. 10). | Model Maker | ☐ | |
| **After Fri Feb 12** Lesson 4 | Case #4: solve all 6 Mystery Patients (p. 11). | Case Solver | ☐ | |
| **Anytime** Bonus | Station Drill with no notes (p. 12): Station 1, then all 5. | Airway Explorer / Station Ace | ☐ | |
| **Anytime** Bonus | 7 days of the Healthy Habits Tracker (p. 14). | Healthy Habits Hero | ☐ | |
| **Anytime** Bonus | Every flashcard set right twice (p. 13). | Flashcard Champ | ☐ | |

## Tournament day: Sat Mar 13

- ☐ 2 sharp pencils (no notes allowed!)
- ☐ A good breakfast and a water bottle
- ☐ Your flashcards for the car ride

- **No notes:** quiz your flashcards every other day.
- Find the aorta first on any heart diagram.
- Right side of the heart = YOUR left.

## CASE CLOSED

This agent earned all 9 badges and is now a **Master Pump Agent**.

Agent _____ Coach or Event Captain _____ Date _____

# My Badges

Color in each badge when you earn it. Earn all 9 to become a **Master Pump Agent**! Your Team Badge Passport has the same badges.



**Pulse Detective**

Case #1: Family Pulse Check

Initials: _____



**Heart Mapper**

Case #2: label all 9 heart parts

Initials: _____



**Blood Tracer**

Case #2: trace the path of blood

Initials: _____



**Model Maker**

Case #3: 5 body models at home

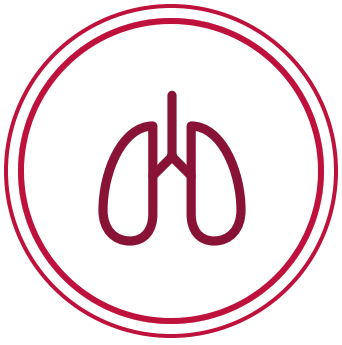
Initials: _____



**Case Solver**

Case #4: solve all 6 patients

Initials: _____



**Airway Explorer**

Put the airway in order

Initials: _____



**Station Ace**

Station Drill with no notes

Initials: _____



**Healthy Habits Hero**

7 days of the Habits Tracker

Initials: _____



**Flashcard Champ**

Every flashcard right twice

Initials: _____

# Answer key

No peeking until you have tried the page!

## Case #2: The Heart Map (p. 9)

1 left ventricle, 2 right atrium, 3 aortic valve, 4 mitral valve, 5 aorta, 6 right ventricle, 7 pulmonary valve, 8 left atrium, 9 tricuspid valve.

**Path:** Body, right atrium, tricuspid valve, right ventricle, pulmonary valve, lungs, left atrium, mitral valve, left ventricle, aortic valve, aorta, body. The right side has oxygen-poor blood. Blood goes to the lungs to drop off carbon dioxide and pick up oxygen. Valves are one-way doors that keep blood moving forward. The left ventricle has the thickest wall because it pumps blood to the whole body.

## Case #1 (p. 8)

Working muscles need more oxygen and make more carbon dioxide, so the heart pumps faster and the lungs breathe faster, together.

## Case #3 (p. 10), sample answers

Trachea: vacuum or dryer hose. Airway tree: broccoli or a bare branch upside down. Spongy lungs: a sponge. Lung filling: a balloon. Blood vessel: a straw or hose. Heart pumping: a squeeze bottle or turkey baster. Valve: a pet door that swings one way.

## Case #4: Mystery Patients (p. 11)

1 Asthma. 2 Heart attack. 3 Sickle cell anemia. 4 COPD. 5 Stroke. 6 Cystic fibrosis. Organs: lungs (airways); heart muscle; blood (red blood cells); lungs; brain; lungs and digestive system. Born with it: patients 3 and 6. Call 911: patients 2 and 5.

## Station Drill (p. 12)

**Station 1:** 1 larynx (voice box), 2 trachea (windpipe), 3 bronchi, 4 bronchioles, 5 alveoli, 6 right lung, 7 left lung, 8 diaphragm. Air path: pharynx, larynx, trachea, bronchi, bronchioles, alveoli. **Station 2:** 1 red blood cells, 2 platelets, 3 white blood cells. **Station 3:** 4 arteries, 5 capillaries, 6 veins. **Station 4:** 7: 22 x 4 = 88 beats per minute. 8: 35 x 4 = 140. 9: Her working muscles needed more oxygen, so her heart pumped faster. **Station 5:** The lungs bring oxygen into the blood and remove carbon dioxide. The heart pumps the oxygen-rich blood to the body and sends oxygen-poor blood back to the lungs.

## Healthy habits (p. 14)

Not smoking: COPD, lung cancer, heart attack, stroke. Exercise and healthy food with less salt: heart attack, stroke, congestive heart failure. Radon test: lung cancer.